WRITING SAMPLE · PROGRAM EVALUATION

Impact Evaluation of a Graduate Program Evaluation Course

A one-group pretest-posttest impact evaluation using paired t-tests and effect sizes to estimate the course's effect on participants' core competencies.

DOCUMENT	**Evaluation report**
AUTHOR	**Ciana Sorrentino Wallace, MPA**
COURSE	**Public Program Evaluation (graduate)**
INSTITUTION	**O'Neill School of Public and Environmental Affairs, Indiana University**
ORIGINAL DATE	**November 2021**

ACADEMIC WORK NOTICE

This report was originally prepared as graduate coursework at Indiana University's O'Neill School of Public and Environmental Affairs, using de-identified class pre- and post-test data provided for instructional purposes. It has been reformatted for portfolio presentation.

No individual student data are reproduced. The design, analysis and conclusions are the author's original work. Prepared for portfolio and professional evaluation purposes.

Introduction

This report measures the overall impact of the Public Program Evaluation course on its graduate student participants through an impact evaluation.

We extract the logic model for the course by reviewing the course syllabus, class objectives, and both mandatory and recommended resources. Using this logic model, we hypothesize, both non-directionally and directionally, that (1) there is a relationship between the program and post-test scores, and (2) the program directly causes an increase in post-test scores.

Design

We employ a quasi-experimental one-group pretest-posttest design: a pre-test measure of the outcome of interest is obtained before the treatment is administered, followed by a post-test on the same measure after treatment. Along with the absence of a control group and non-random participant selection, this design means that all participants receive the same treatment and that the dependent variable is assessed in a linear order. The effect of the treatment can therefore be estimated as the difference between pre-test and post-test scores.

The primary quantitative measure is each student's total score on the course components tested in the pre- and post-tests, which shows how much the program worked from the beginning to the end of the course. The outcomes of interest are the quantified share of class objectives demonstrated on the post-test at the point of program completion. In essence, we evaluate whether the program reliably increased knowledge and behavioral core competencies.

Methodology and findings

Statistical significance

To identify effects, we first analyze the pre- and post-test data separately. The results are then paired for a t-test of the change in test scores across the 37 participants. Because it is reasonable to hypothesize that results fall into only one tail of the distribution, a one-tailed t-test is appropriate; statistical significance is achieved if the mean post-test score is higher than the mean pre-test score. Both the one-tailed and two-tailed paired t-tests return very small p-values ($p < 0.001$), so we reject the null hypothesis: the change in scores is statistically significant.

Effect size

To understand the strength of the difference between the two test scores, we calculate the effect size for the paired t-test: the difference between the mean post-test and mean pre-test scores, divided by the standard deviation of the pre-test scores. This magnitude represents the number of standard deviations of difference attributable to the program. An effect size of 1.8 indicates a large effect of the program on participants.

Competencies with the most growth

Examining the average difference by question shows which core competencies the program affected most. The largest gains were on three items:

- Q7: identifying the counterfactual ("What would have happened in the absence of the program is called the ___").
- Q8: describing selection bias and giving an example of it in a hypothetical or actual program evaluation.
- Q13: listing three distinct types of validity of measures.

On average, participants improved by one point on each of these items between the pre- and post-tests, indicating that the program had its greatest effect on these areas of understanding.

Limitations and recommendations

Threats to internal validity

Pre-/post-test designs pose several threats to internal validity, making it more difficult to infer causality. Testing is one threat: the pre-test itself may have influenced students' knowledge of the core competencies. Given the time between the pre- and post-tests, co-occurring events may also bias the results; students may be taking other classes or working in a profession that influenced their knowledge. Instrumentation is a further threat, as burnout or fatigue over the course may have made students less careful with their post-test responses.

Regression to the mean offers an alternative explanation for the change in scores among participants with outlier pre-test scores, who would show smaller gains even if the training had an effect. To address these limitations, an additional pre-test measure could reduce the effects of maturation and regression to the mean, ideally one tied more closely to behavioral rather than solely knowledge-based core competencies.

Testing behavioral competencies

Despite the medium-sized, non-random sample, the large effect size indicates that the difference is both statistically and practically significant, and the data are amenable to statistical analysis. To test the behavioral dimension of the core competencies with the available data, we also tested differences in time taken to complete the tests. These results were not statistically significant (all p-values greater than 0.05), which strengthens the case for adding a pre- and post-test measure directly related to behavioral competencies to better capture the outcomes and impacts in the program's logic model. The study would also benefit from replication with a larger sample.